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The organizers of the Summer School
Structures in Lie Representation Theory
August 9 - 22, 2009
Jacobs University Bremen, Germany

Dear Organizers of the Summer School,

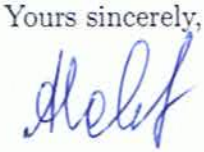
I am writing to support the funding application of my former PhD student Lucy Gow. Lucy successfully completed her PhD project on Lie superalgebras and Yangians in the School of Mathematics and Statistics, University of Sydney, 2008. Lucy's thesis is concerned with the study of quantum algebras called Yangians which are associated with the general and special linear Lie superalgebras. Such algebras emerged in 1980s in the works of mathematical physicists and have since found many applications in and connections with theoretical physics, algebraic combinatorics, geometry and low-dimensional topology.

The main results of Lucy's thesis are a proof of the Poincaré–Birkhoff–Witt theorem for the Yangian for the general linear Lie superalgebra, a new Drinfeld-type presentation of the Yangian, a new expression for the quantum Berezinian and a proof of Nazarov's conjecture (1991) that the coefficients of the quantum Berezinian generate the center of the super Yangian. The results thus provide a comprehensive description of the algebraic structure of the super Yangians and open the possibility to explore their representation theory and connections with the emerging theory of finite W-algebras.

The results of the thesis have been published in two papers in the *Czechoslovak Journal of Physics* and the *Communications in Mathematical Physics.*

Lucy is actively working on research projects on representations theory of Yangian-type algebras and the applications of super Yangians in physics. It would be a valuable experience for her to take part in the Summer School in Bremen and would be greatly beneficial for her research.

Yours sincerely,



A. I. Molev